

opportunity cost components are more attributable to investment managers than to the trading party.

The trading cost or arrival cost component is:

$$\begin{aligned} \text{Arrival Cost}_{\$} &= \sum s_j p_j - \left(\sum s_j \right) P_0 & (3.9) \\ S, s_j &> 0 \text{ for buys} \\ S, s_j &< 0 \text{ for sells} \end{aligned}$$

In basis points this expression is:

$$\text{Arrival Cost}_{bp} = \frac{\sum s_j p_j - (\sum s_j) P_0}{(\sum s_j) P_0} \cdot 10^4 \quad (3.10)$$

In general, arrival costs can be simplified as follows:

$$\text{Arrival Cost}_{bp} = \text{Side} \cdot \frac{P_{avg} - P_0}{P_0} \cdot 10^4 \quad (3.11)$$

where,

$$\text{Side} = \begin{cases} 1 & \text{if Buy} \\ -1 & \text{if Sell} \end{cases}$$

EVALUATING PERFORMANCE

In this section we describe various techniques to evaluate performance (note: we will use the profit and loss (PnL) terminology). These methods can be used to evaluate and compare trade quality for a single stock or basket of trades, as well as performance across traders, brokers, or algorithms. It can also serve as the basis for universe comparisons. In the following section we provide non-parametric statistical techniques that are being used to compare algorithmic performance.

Techniques that will be discussed in this section include: market or index adjusted cost, benchmark comparisons, various volume weighted average price (VWAP), participation weighted average price (PWP), relative performance measure (RPM), and z-score statistical measures.

Trading Price Performance

Trading price performance or simply trading PnL is identical to the trading cost component above and is measured as the difference between